Lab 5.2 source pack

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

DES-01-rev2.md

DES-01 Pump design note revision 2
Fictional training reference. Effective 2026-08-01. Supersedes revision 1.
Clause 2 Duty
Nominal delivery is 50 m3/h at 30 m head.
Clause 3 Classroom acceptance
The teaching acceptance band is plus or minus 5 percent of nominal delivery, hence 47.5 to 52.5 m3/h. Record three readings at agreed duty, instrument identity and witness. This is not a published engineering standard.
Clause 4 Materials
Wetted materials remain TBD pending fluid identification and compatibility review.
Clause 5 Release
Responsible engineer review is required before any real-world use.

OM-01.md

OM-01 Dosing skid information revision 1
Fictional training reference only. Clause 1: four elements are T-1, DP-1, CV-1 and FT-1. Clause 2: the training case has annual FT-1 calibration and effectiveness has not been demonstrated. Clause 3: abnormal readings need investigation; no cause is established by this manual. Clause 4: no chemical identity or operating instructions are supplied. Refer operational decisions to approved site procedures.

report-template.md

Weekly engineering progress report
Reporting period | Author | Version | Draft status
Executive summary
Completed work with evidence
Open actions and owners
Risks and blockers
Decisions required
Next period plan
Sources and unresolved assumptions
Reviewer and release decision

rating-scales.txt

IES TRAINING FMEA SCALE revision 1
This is a simplified classroom scale, not an industry or company standard. Use integer ratings 1-10.
Severity: 1 negligible training impact; 4 local rework; 7 major functional loss; 10 potentially catastrophic consequence. Intermediate ratings need written justification.
Occurrence: 1 remote; 4 occasional; 7 frequent; 10 persistent. These are qualitative classroom anchors, not probabilities.
Detection: 1 almost certain detection before the effect; 4 likely; 7 unlikely; 10 no effective detection. A larger D means poorer detectability.
RPN = S x O x D. Review high-severity scenarios separately even when RPN is lower. Proposed actions do

not reduce ratings until their effectiveness has been verified.

system-description.txt

SYS-01 revision 1 | fictional chemical dosing skid
Function: deliver a controlled dose to a training water line. Elements: tank T-1; metering pump DP-1; check valve CV-1; flow sensor FT-1. Current controls: tank level alarm checked each shift; weekly pump check; monthly valve inspection; annual sensor calibration. Control effectiveness is unknown. Failure candidates: no flow, excessive flow, reverse flow, sensor drift. Fluid identity, real exposure limits and approved materials are outside the case. Treat all consequences and ratings as workshop hypotheses requiring engineering review.

fmea-template.csv

id: F01; function: Deliver controlled dose; failure_mode: No flow; effect: Underdose; cause: Pump failure; current_control: Weekly check; S: 8; O: 4; D: 6; RPN: ; action: Review redundancy and alarm evidence; owner: Engineer A; due_date: 2026-09-12; evidence: SYS-01 rev1; status: Proposed

id: F02; function: Prevent reverse flow; failure_mode: Reverse flow; effect: Cross contamination; cause: Valve stuck; current_control: Monthly inspection; S: 9; O: 3; D: 7; RPN: ; action: Review backflow protection; owner: Engineer B; due_date: 2026-09-12; evidence: SYS-01 rev1; status: Proposed

id: F03; function: Measure dose; failure_mode: Sensor drift; effect: Incorrect adjustment; cause: Calibration drift; current_control: Annual calibration; S: 7; O: 5; D: 5; RPN: ; action: Review calibration frequency; owner: Engineer C; due_date: 2026-09-15; evidence: SYS-01 rev1; status: Proposed

handoff-template.md

Agent handoff record
Question | Retrieval date | Source title | Version | Clause or page | Exact supported statement | Source URL | Access scope | Uncertainty | Target document section | Reviewer decision